Project Name: Project Amaze Amaze Amaze

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Project Design Document

1. Design Intent

- *We are building a simplified robotic version of Rocky (from Project Hail Mary)*
- *It's a 5-limb spider-like creature that communicates through gestures and moves using its limbs.*
- *The main target audience is just robotics and movie enthusiasts, or anyone in general who enjoys interacting or playing with robots.*

2. Inspiration & References

Idea Inspiration

- *Project Hail Mary the movie, primary inspiration for Rocky's five-limbed body and gestural communication style*
- *Rocky in the book moves using its back limbs for locomotion while the front limbs are used for interaction and expression*
- *We mainly gained a basic understanding of how Rocky functions, its key features, and movements we can showcase.*

3. User Journey

Step-by-step experience of the user:

1. *User starts with opening the app created to control Rocky.*
2. *Various options appear from which the User can select what they want Rocky to do.*
3. *User interacts with app by choosing the following options:*
    a. *Thumbs down action*
    b. *Fist bump action*
    c. *Moving forward*
    d. *Moving backward*
4. *Rocky responds according to the chosen option.*

4. Success Criteria

Define measurable success:

Function:

- *All limbs are operating efficiently.*
- *Ideal body balance is achieved.*
- *System responds to the option selected on the app.*

User experience (easy to use, intuitive)

- *The app UI is easy to navigate and operate.*
- *It is visually appealing.*
- *Should be fun to interact with.*

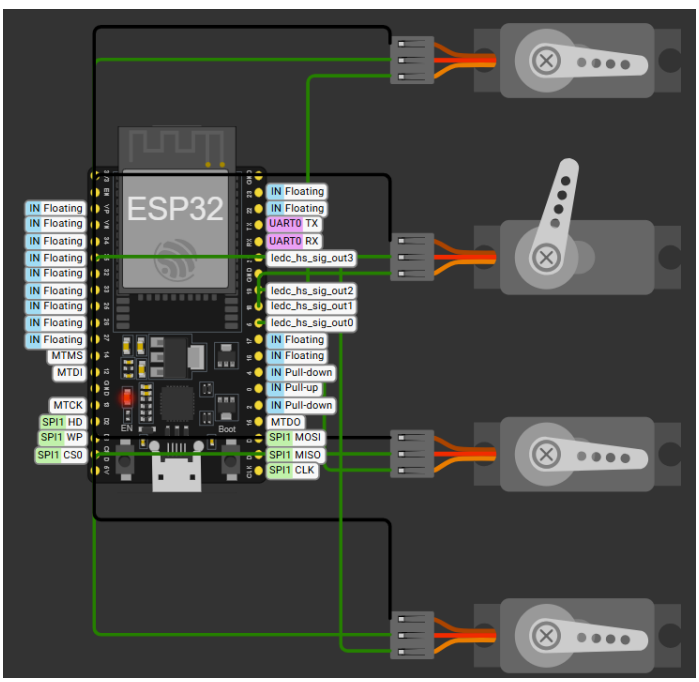
5. System Overview

- *We are making a robotic version of Rocky from the movie Project Hail Mary. In our project, Rocky has 5 limbs – 4 limbs for movement and 1 limb for hand gestures.*
- *We will be using 1 servo motor each for 4 limbs and then 3 servo motors for the 5th limb (the hand).*
- *When the user chooses to press the forward option in the app, each of the 4 limbs meant for movement, will move forward one by one, with the help of the respective servo motors, hence making the whole body move forward. This same logic is applicable for moving backward as well.*

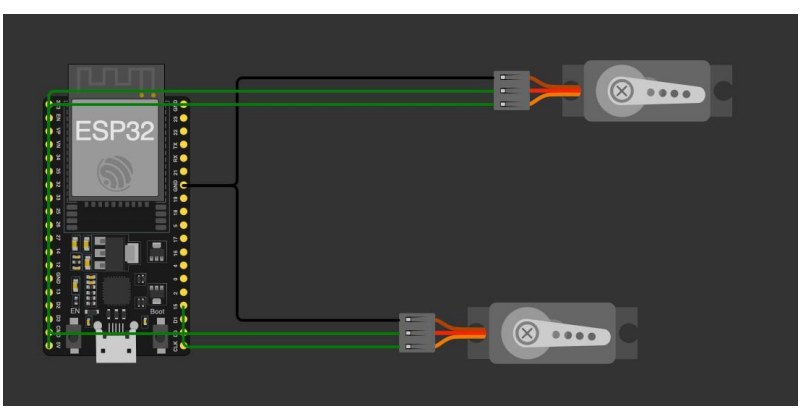
- *When the user chooses an interactive hand gesture like “fist bump”, Rocky moves the elbow of his front limb to 90 degrees, indicating a fist gesture.*
- *Likewise, when the user chooses the “thumbs down” gesture, Rocky moves the elbow of his front limb to 90 degrees and then one part of his hand goes down by 90 degrees, indicating a thumbs down gesture.*
- *We will be using 6 servo motors in total, ESP32, power supply module, motor driver.*
- *Model will be made with cardboard and paper. Skeleton will be made using laser cutting.*

6. High-Level Block Diagram

4 Limbs (legs).



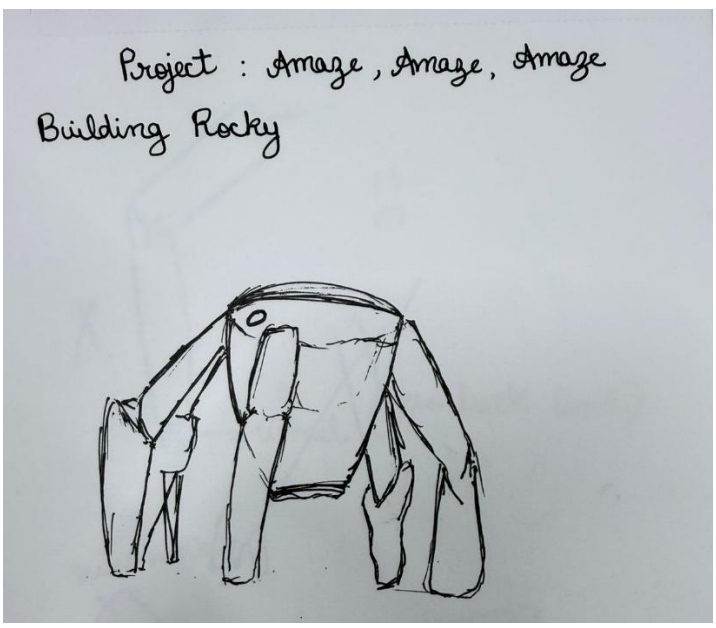
Front Limb (elbow and thumb)

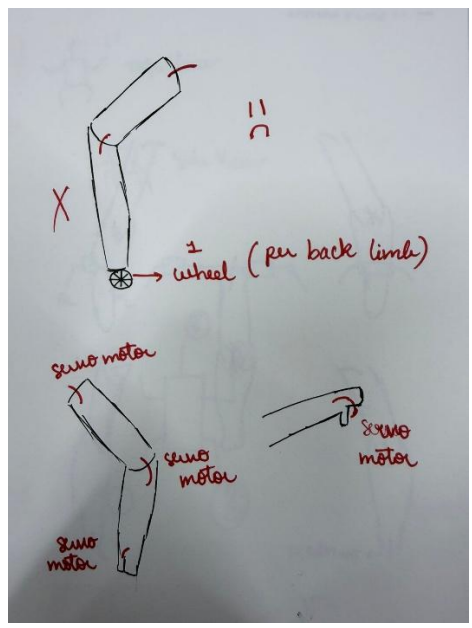


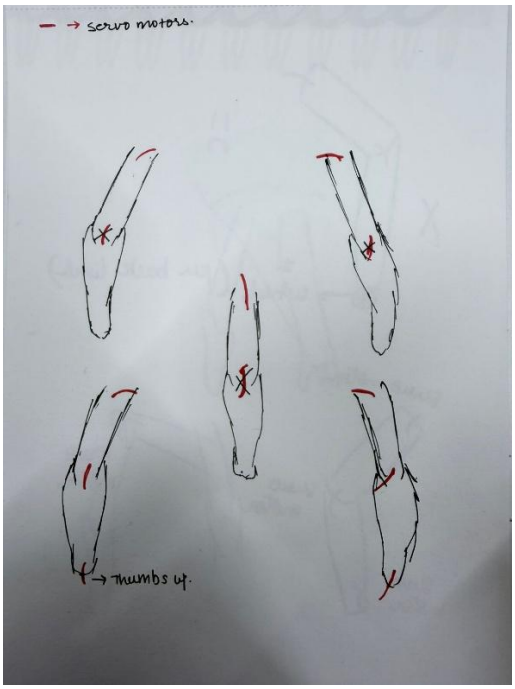
7. Input / Output Mapping

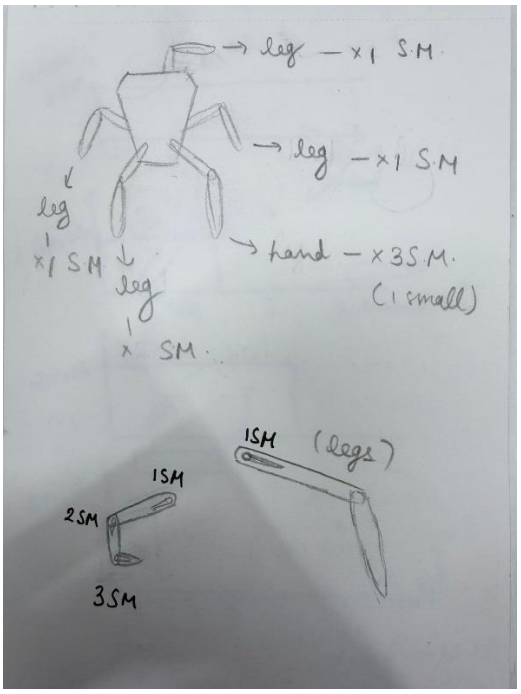
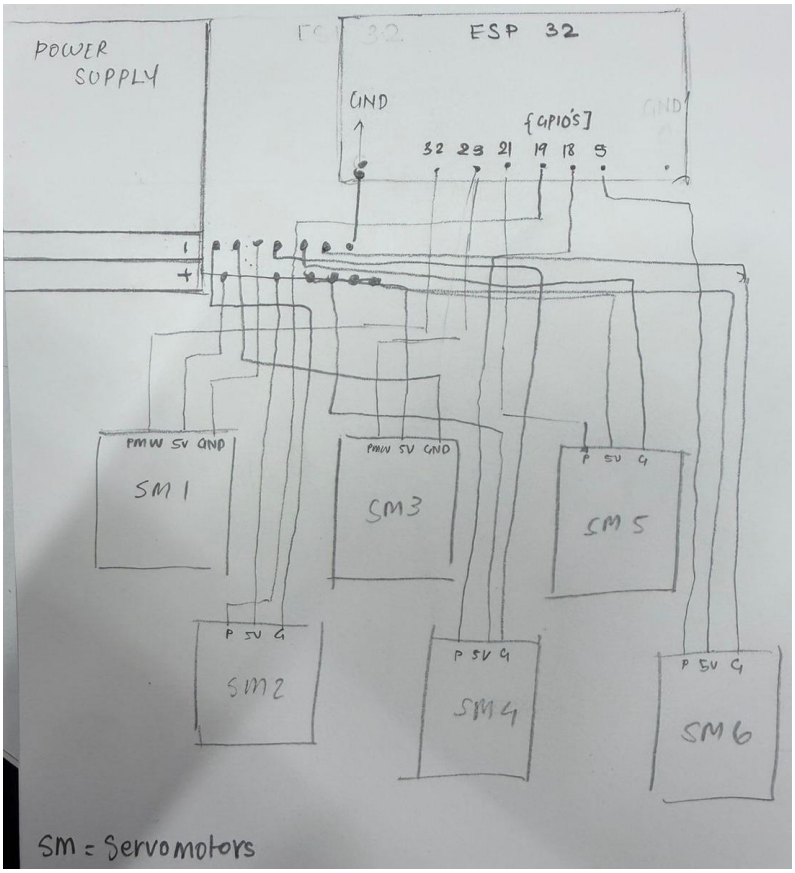
Input (app)	Process	Output
Forward Command	**Systems receive message**	**Servo motor executes functions**
Backwards Command	**Systems receive message**	**Servo motor executes functions**
Thumbs down command	**Systems receive message**	**Servo motor executes functions**
Fist Bump Command	**Systems receive message**	**Servo motor executes functions**
Reset Command	**Systems receive message**	**Servo motor executes functions**

8. Sketches & Concepts









9. Final System Architecture (Labelled)